COMMUTATION WITH COUNTABLE SUMS UPGRADES TO THE COLIMIT OF A COUNTABLE SYSTEM OF COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 20331-20345 (canonical tag 07LR) and the counted proof is the proof environment at source lines 20347-20376, which immediately follows it and is its own proof.

Editorial note. The proof displays the long exact sequence in the author's own Xy-pic markup, which is compiled from that markup with the shared preamble.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which includes the appended environment definitions; the counted text does not use them, and the span is taken whole so that the file version is unambiguous.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key and by this note. The counted statement and proof cite no bibliography entry, so this card needs no bibliography.

Allowed standard prerequisites (editorial).

- For a system of complexes indexed by the natural numbers the termwise colimit fits into a short exact sequence from the countable sum to itself given by the shifts of the transition maps. used at proof lines 20348-20353. Displays the sequence whose associated triangle expresses the colimit in terms of the countable sum.
- The shift of a morphism of complexes defines a distinguished triangle in the derived category. invoked at proof lines 20353-20355. Turns the displayed short exact sequence into a distinguished triangle, which is what allows a long exact sequence to be written.
- The functor represented by a fixed object of the derived category is homological, so a distinguished triangle yields a long exact sequence of abelian groups. invoked at proof lines 20356-20357. Produces the displayed long exact sequence from which the conclusion is read.

Source reference. Original lemma 0152: sources/stacks-chow/tags.txt, source lines 1493–1493. The reference converts the displayed short exact sequence of complexes into a distinguished triangle, used at proof lines 20353-20355; the long exact sequence and the conclusion are inside the counted span.

Source reference. Original lemma 0149: sources/stacks-chow/tags.txt, source lines 1464–1464. The reference is the homological property of the represented functor, invoked at proof lines 20356-20357 to obtain the long exact sequence; the reading of the exactness that follows is inside the counted span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Let $K \in D(R)$ be an object such that for every countable set of objects $E_n \in D(R)$ the canonical map*

$$\bigoplus \mathrm{Hom}_{D(R)}(K, E_n) \longrightarrow \mathrm{Hom}_{D(R)}(K, \bigoplus E_n)$$

is a bijection. Then, given any system $L_n^\bullet$ of complexes over $\mathbf{N}$ we have that

$$\mathrm{colim}\, \mathrm{Hom}_{D(R)}(K, L_n^\bullet) \longrightarrow \mathrm{Hom}_{D(R)}(K, L^\bullet)$$

is a bijection, where $L^\bullet$ is the termwise colimit, i.e., $L^m = \mathrm{colim}\, L_n^m$ for all $m \in \mathbf{Z}$.

Proof. Consider the short exact sequence of complexes

$$0 \to \bigoplus L_n^\bullet \to \bigoplus L_n^\bullet \to L^\bullet \to 0$$

where the first map is given by $1 - t_n$ in degree n where $t_n : L_n^\bullet \to L_{n+1}^\bullet$ is the transition map. By Derived Categories, Lemma 0152 this is a distinguished triangle in $D(R)$. Apply the homological functor $\mathrm{Hom}_{D(R)}(K, -)$, see Derived Categories, Lemma 0149. Thus a long exact cohomology sequence

$$\begin{array}{ccccc} & & \ldots & \longrightarrow & \mathrm{Hom}_{D(R)}(K, \mathrm{colim}\, L_n^\bullet[-1]) \\ & \swarrow & & & \\ \mathrm{Hom}_{D(R)}(K, \bigoplus L_n^\bullet) & \longrightarrow & \mathrm{Hom}_{D(R)}(K, \bigoplus L_n^\bullet) & \longrightarrow & \mathrm{Hom}_{D(R)}(K, \mathrm{colim}\, L_n^\bullet) \\ & \swarrow & & & \\ \mathrm{Hom}_{D(R)}(K, \bigoplus L_n^\bullet[1]) & \longrightarrow & \ldots & & \end{array}$$

Since we have assumed that $\mathrm{Hom}_{D(R)}(K, \bigoplus L_n^\bullet)$ is equal to $\bigoplus \mathrm{Hom}_{D(R)}(K, L_n^\bullet)$ we see that the first map on every row of the diagram is injective (by the explicit description of this map as the sum of the maps induced by $1 - t_n$). Hence we conclude that $\mathrm{Hom}_{D(R)}(K, \mathrm{colim}\, L_n^\bullet)$ is the cokernel of the first map of the middle row in the diagram above which is what we had to show. $\square$